

CHE 321 - Exam 2 - Sample 3

Exam2fun.m

```
1  function h = Exam2fun(V,y)
2  UA = 80;
3  Cpcool = 10;
4  mcool = 50;
5  CpA = 10;
6  CpB = 10;
7  CpC = 30;
8  CpD = 20;
9  k1A = 40;
10 k2C = 2;
11 deltaHa = -15000;
12 deltaHc = -10000;
13 T0 = 300;
14 CT0 = 0.2;
15
16
17 FA = y(1);
18 FB = y(2);
19 FC = y(3);
20 FD = y(4);
21 T = y(5);
22 Ta = y(6);
23
24 FT = FA + FB + FC + FD;
25
26 CA = CT0 * (FA/FT)*(T0/T);
27 CB = CT0 * (FB/FT)*(T0/T);
28 CC = CT0 * (FC/FT)*(T0/T);
29
30 Sum = FA *CpA + FB * CpB + FC * CpC + FD * CpD;
31
32 ratea = k1A * CA * CB^2;
33 ratec = k2C * CA * CC;
34
35 Qg = (ratea * deltaHa) + (ratec * deltaHc);
36 Qr = UA*(T - Ta);
37
38 h1 = -ratea - ratec;
39 h2 = -2 * ratea;
40 h3 = ratea - ratec;
41 h4 = ratec;
42 h5 = (Qg - Qr)/(Sum);
43 h6 = (UA * (T-Ta))/(mcool * Cpcool);
44
45
46 h = [h1,h2,h3,h4,h5,h6]';
```

Exam2.m

```
1  clc;
2  clear all;
3
4  FA0 = 5;
5  FB0 = 10;
6  FC0 = 0;
7  FD0 = 0;
8  T0 = 300;
9  Ta0 = 325;
10 CT0 = 0.2;
11
12 y0 = [FA0, FB0, FC0, FD0, T0, Ta0];
13 Vspan = [0 10];
14
15 [V, y] = ode45(@Exam2fun, Vspan, y0);
16
17 FA = y(:,1);
18 FB = y(:,2);
19 FC = y(:,3);
20 FD = y(:,4);
21 T = y(:,5);
22 Ta = y(:,6);
23
24 FT = FA + FB + FC + FD;
25 CA = CT0 * (FA./FT) * (T0/T);
26 CB = CT0 * (FB./FT) * (T0/T);
27 CC = CT0 * (FC./FT) * (T0/T);
28 CD = CT0 * (FD./FT) * (T0/T);
29
30
31
32 figure(1);
33 plot(V, FA, V, FB, V, FC, V, FD);
34 xlabel('Volume');
35 ylabel('Molar Flow Rates');
36 legend('FA', 'FB', 'FC', 'FD');
37 title('ODE Solution Results');
38 grid on;
39
40 figure(2);
41 plot(V, CA, V, CB, V, CC, V, CD);
42 xlabel('Volume');
43 ylabel('Concentrations');
44 legend('CA', 'CB', 'CC', 'CD');
45 title('ODE Solution Results');
46 grid on;
47
48
49 figure(3);
50 plot(V, T, V, Ta);
51 xlabel('Volume');
52 ylabel('Temperature');
53 legend('T', 'Ta');
54 title('ODE Solution Results');
55 grid on;
```